Zero-shot downstream evaluation

Final checkpoints | 14 models | 10 tasks | Scores in percent

Size	Model	PPL (down)	LAMBADA	PIQA	WinoGrande	ARC-E	ARC-C	SciQ	LogiQA	BoolQ	HellaSwag	RACE-M	Avg
700M	Transformer	9.72	30.60	68.61	54.30	**60.31**	29.78	88.30	23.35	59.94	45.55	39.42	50.02
	AttnRes	9.67	31.22	68.44	51.54	58.59	**32.34**	87.50	23.35	**61.41**	45.81	40.95	50.11
	MUDDFormer	9.38	33.38	68.61	53.91	57.70	**32.34**	**89.90**	**23.96**	60.58	47.05	40.53	50.80
	M3 (dd40)	**9.31**	**35.82**	**68.88**	52.96	57.79	30.80	89.00	22.27	61.22	**48.08**	41.64	50.85
	M3 (dd80)	**9.31**	34.29	68.66	**55.25**	59.93	31.57	89.60	22.27	61.25	47.62	**41.99**	**51.24**
1.4B	Transformer	8.33	38.29	71.38	58.64	**64.65**	35.49	90.40	20.74	61.04	54.56	42.97	53.82
	AttnRes	8.21	39.39	**71.65**	**59.04**	62.71	**37.63**	91.00	**24.42**	62.42	54.71	42.27	54.52
	M3 (dd26)	**8.07**	**40.97**	70.84	58.88	63.76	35.92	**91.70**	22.89	**64.28**	**55.92**	**46.17**	**55.13**
1.8B A0.38B	Transformer	9.73	27.38	67.63	52.17	**56.02**	30.12	84.80	21.81	**61.44**	44.33	35.93	48.16
	AttnRes	11.12	25.07	63.82	52.96	48.91	26.54	83.80	22.89	60.37	38.74	36.84	45.99
	MUDDFormer	9.97	28.90	66.59	52.49	53.91	29.78	86.40	**24.42**	58.50	43.03	**39.14**	48.32
	M3 (dd64)	**9.67**	**30.33**	**67.68**	**53.20**	54.76	**30.29**	**87.50**	23.66	55.78	**44.84**	37.81	**48.58**
4.6B A0.89B	Transformer	8.07	35.51	**72.74**	56.99	63.05	**36.26**	**90.00**	**20.28**	55.41	54.16	42.41	52.68
	M3 (dd96)	**8.02**	**38.00**	71.06	**57.06**	**64.23**	36.01	89.80	19.35	**59.85**	**54.56**	**43.25**	**53.32**

Metrics: PIQA, ARC-E, ARC-C and HellaSwag use acc_norm; the other six tasks use accuracy. Avg is the unweighted mean of these ten scores.
PPL is from the original held-out language-model validation set, is identical across shot settings, and is excluded from Avg. RACE-H is excluded.
AttnRes denotes Block Attention Residuals (4 Transformer layers per block). M3 denotes dd; group counts are shown. The 1.4B dd26 uses no PrePostDANorm.
MoE sizes are rounded total / active parameters; A denotes active parameters. Bold indicates the best displayed value within a size, including rounded ties.
Protocol: LAMBADA standard; RACE-M evaluated per question; fixed few-shot seed 47. Context cap 4096, no prompt truncation and no RoPE modification.
Training context is 2048. In 5-shot, 170 RACE-M questions and one question each in LogiQA and BoolQ exceed 2048; the longest prompt plus answer is 2557 tokens.
Single training seed; no statistical-significance claim. Pretraining/benchmark near-duplicate decontamination has not been audited.
Inference: Dense BF16 with causal SDPA; MoE FP32 parameters with BF16 autocast. Large-MoE diagnostics showed shape-dependent logits; fixed-shape causality passed.

Five-shot downstream evaluation

Final checkpoints | 14 models | 10 tasks | Scores in percent

Size	Model	PPL (down)	LAMBADA	PIQA	WinoGrande	ARC-E	ARC-C	SciQ	LogiQA	BoolQ	HellaSwag	RACE-M	Avg
	Transformer	9.72	29.90	68.72	55.33	65.70	32.85	90.20	22.43	**62.26**	45.71	41.50	51.46
	AttnRes	9.67	30.04	67.41	52.80	66.37	34.04	90.60	20.58	56.73	45.37	41.57	50.55
700M	MUDDFormer	9.38	30.56	68.66	54.06	68.48	34.47	90.40	**25.04**	58.65	47.06	42.20	51.96
	M3 (dd40)	**9.31**	33.05	68.88	54.85	67.76	**34.90**	**92.10**	23.04	59.45	**47.99**	**44.01**	**52.60**
	M3 (dd80)	**9.31**	**33.71**	**69.04**	**55.96**	**68.69**	**34.90**	91.80	23.35	53.15	47.84	43.38	52.18
	Transformer	8.33	36.25	**72.25**	59.35	72.39	39.16	93.90	22.43	62.78	54.67	45.26	55.85
1.4B	AttnRes	8.21	40.40	71.55	58.64	72.35	**40.36**	93.40	**23.50**	**62.97**	54.72	44.85	56.27
	M3 (dd26)	**8.07**	**40.42**	71.65	**61.17**	**72.90**	39.51	**94.60**	21.81	59.69	**56.28**	**47.14**	**56.52**
	Transformer	9.73	26.14	**69.04**	51.85	**65.78**	**33.28**	88.60	25.81	56.82	44.19	36.00	49.75
1.8B A0.38B	AttnRes	11.12	24.65	64.64	53.59	56.40	28.24	85.80	22.58	43.46	38.07	36.00	45.34
	MUDDFormer	9.97	24.53	67.90	**54.30**	63.05	32.76	**89.20**	25.81	**59.36**	42.87	**39.35**	**49.91**
	M3 (dd64)	**9.67**	**26.57**	67.57	51.38	64.65	32.34	89.10	**26.73**	53.06	**44.84**	38.65	49.49
4.6B A0.89B	Transformer	8.07	**36.99**	**73.01**	**56.99**	**71.97**	**40.27**	92.30	21.51	60.24	54.24	43.31	55.08
	M3 (dd96)	**8.02**	36.02	71.60	56.67	**71.97**	38.74	**93.30**	**23.96**	**61.19**	**55.14**	**46.03**	**55.46**

Metrics: PIQA, ARC-E, ARC-C and HellaSwag use acc_norm; the other six tasks use accuracy. Avg is the unweighted mean of these ten scores.
PPL is from the original held-out language-model validation set, is identical across shot settings, and is excluded from Avg. RACE-H is excluded.
AttnRes denotes Block Attention Residuals (4 Transformer layers per block). M3 denotes dd; group counts are shown. The 1.4B dd26 uses no PrePostDANorm.
MoE sizes are rounded total / active parameters; A denotes active parameters. Bold indicates the best displayed value within a size, including rounded ties.
Protocol: LAMBADA standard; RACE-M evaluated per question; fixed few-shot seed 47. Context cap 4096, no prompt truncation and no RoPE modification.
Training context is 2048. In 5-shot, 170 RACE-M questions and one question each in LogiQA and BoolQ exceed 2048; the longest prompt plus answer is 2557 tokens.
Single training seed; no statistical-significance claim. Pretraining/benchmark near-duplicate decontamination has not been audited.
Inference: Dense BF16 with causal SDPA; MoE FP32 parameters with BF16 autocast. Large-MoE diagnostics showed shape-dependent logits; fixed-shape causality passed.

Source: downstream-eval-20261028/results-{0,5}shot.csv | Evaluation completed 2026-10-28 18:17 CST